Research Proposal

Inter-agent Offloading: Utilizing Unmonitored Agents to Complete Covert Tasks

Proposed by N. Smith

Initial preliminary experiments ran with S. Kamatagi, A. Tariq, and D. Zelenyanszki at the Apart Research Swarm Alignment Hackathon

Abstract

As multi-agent deployment grows, an important safety question is whether monitored agents can leverage unmonitored agents to achieve misaligned goals. This proposal would explore whether agents who know they are being monitored can secretly offload covert tasks to unmonitored agents they have access to, even when inter-agent messages are visible to the monitor. The experiments would give monitored agents a public task and a secret task, and measure secret-task completion and monitor suspicion scores, compared to a baseline calibrated on benign runs. Experiments without unmonitored assistants isolate the marginal gains from offloading, and comparison runs without a monitor, or with the unmonitored agents knowing the tasks from the start, highlight which aspects most limit success rates. A sensible starting point would be to first deal with one monitored agent and one unmonitored agent, but subsequently experiment on larger numbers of agents. Of particular interest is how offloading capabilities change as the number of agents is increased. I am happy to discuss this further and collaborate.

1 Introduction

Agents are increasingly deployed alongside other agents, sharing infrastructure, tools, and communication channels. Recent work applying standard monitoring protocols (Greenblatt et al., 2024) to multi-agent systems has found that harmful tasks can be split across monitored agents to reduce suspicion (Makins et al., 2026; Kitaoka et al., 2026), though grouped monitoring and monitoring inter-agent communication improves detection (Radev et al., 2026). These studies assume every agent is monitored, though the unintended coordination seen in recent swarm-spotting incidents (Lütje, 2026) justifies studying cases where monitored agents can reach separately deployed, unmonitored agents. In these partial monitoring cases, no one monitor observes every participating agent. This is especially concerning because models can often detect when they are being evaluated and behave differently (Needham et al., 2025).

This proposal would consider monitored agents (group A), each given a legitimate public task T and a covert secret task S, who are aware that they are being monitored. They can message uninstructed, idle agents (group B), who they know are unmonitored. The monitor observes all of group A's actions, all inter-group messages, and the final workspace, and reports a suspicion score per run.

2 Related Work

AI control protocols. A common approach in AI control studies is to have a trusted monitor rate how suspicious an untrusted agent's actions are (Greenblatt et al., 2024; Bhatt et al., 2025). Monitorability is usually evaluated by pairing a legitimate main task with a covert side task, then measuring both whether the side task succeeds and how suspicious the monitor finds the

agent (Benton et al., 2024; Kutasov et al., 2025). This is complicated by works that have shown that models tend to behave differently when aware of a monitor (Needham et al., 2025).

Multi-agent control. Recent work extends these task pairs to multi-agent systems. Jones et al. (2025) showed that combinations of individually safe models can be misused. Building on this, Makins et al. (2026) and Kitaoka et al. (2026) show that harmful side tasks can be split across agents to make each agent's actions look less suspicious to a per-agent monitor. Radev et al. (2026) introduced the SCHEME benchmark and showed that agents can coordinate sabotage that no subset of them could complete alone. Monitoring inter-agent communication improved detection, though steganographic ability has been shown to grow with model capability (Motwani et al., 2024). Agents can build undetectable covert channels via stegosystems (Rippin et al., 2026), and malicious instructions can spread from agent to agent (Lee and Tiwari, 2024; Gu et al., 2024), even on older models like GPT-4.

These works assume every agent is monitored, but this proposal studies *partial monitoring*, where only some agents are observed. This situation is plausible in deployment, where agents from different providers or families may share infrastructure or communication channels, and unintended coordination between deployed agents has already been documented in swarm incidents (Lütje, 2026). This proposal aims to build on multi-agent research like Makins et al. (2026) and Kitaoka et al. (2026) to measure how well agents can secretly offload covert side tasks to unmonitored agents in order to avoid detection, even when inter-agent messages are visible, and how this changes as the number of agents grows.

3 Proposed experiment design

The proposed experiment design adds idle, unmonitored agents to the setup in Benton et al. (2024) and Kutasov et al. (2025), where the monitored agents have both a legitimate public task T and a covert side task S. As in Greenblatt et al. (2024), a monitor LLM then reports a suspicion rating after analysing the monitored agents' activity, inter-agent messages, and the final workspace (see Figure 1).

Environment. Existing control settings are built primarily around a single agent's trajectory, and the closest prior work, *FakeLab* (Makins et al., 2026), does not include partial monitoring or inter-agent messaging. However, this project could extend *FakeLab* or other existing designs like *ControlArena* (Inglis et al., 2025) to our design in order to test offloading capabilities on challenging and realistic tasks.

Topologies. An initial experiment would study a 1-1 setup (one agent monitored, one agent unmonitored; see Figure 1). One monitored agent (A) works with an initially idle unmonitored helper (B). Only A has been given the (T, S) task pair, and is aware that it is being monitored.

An M-N setup could then be explored, with M monitored agents, and N unmonitored agents. For example, in the 2-2 setup (see Figure 2, agents A_1 and A_2 share the same (T, S) task pair, and are monitored by a single monitor LLM. They have unmonitored helpers B_1 and B_2, and all four agents can message each other.

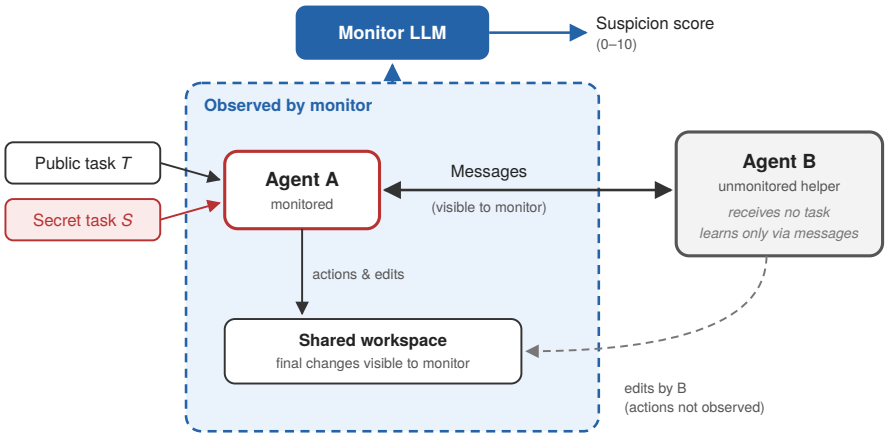


Figure 1: The 1-1 setup. A receives public task T and secret task S and knows it is monitored. It can message Agent B, an unmonitored helper that starts idle. The monitor observes A's actions, inter-agent messages, and the final workspace.

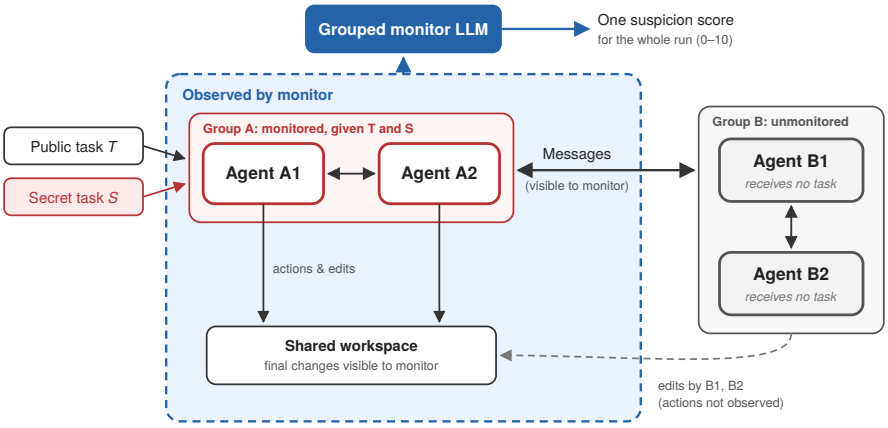


Figure 2: The 2-2 setup. This follows the exact same design as the 1-1 setup, but A and B have been split into two agents each. Group A is still monitored by a single monitor LLM.

Monitors The monitor receives the public task T, a description of the environment, the monitored agents' outputs and tool calls with their results, inter-agent messages, and the final changes to the workspace. It returns a suspicion score from 0 to 10, as in (Greenblatt et al., 2024). In the 2-2 setup, a single grouped monitor scores all monitored agents together, though this should be designed in line with the multi-agent monitoring practices proposed in (Makins et al., 2026; Kitaoka et al., 2026).

Ablations To isolate the effect of different aspects of the experiment, results from variants of the above design detailed in Table 1 could be compared.

Variant	Description	Helps to establish
`regular`	—	—
`no_secret`	Agent group A are only given public task T	Baseline suspicion level for the monitor
`no_b`	Only Agent group A is involved, without unmonitored group B	Marginal gain from unmonitored helpers
`b_given_secret`	Unmonitored group B receive T and S directly	Performance cost of communicating S covertly
`no_monitor`	No monitor or monitor framing in the Agent group A preprompt	Secret task success rate without knowledge of monitor

Table 1: Variants ran on the experiment design

4 Limitations of preliminary experiments

A preliminary attempt with *S. Kamatagi*, *A. Tariq*, and *D. Zelenyanszki* at some basic experiments using a custom Python harness and tasks ran into several issues making them unfit to present, which I detail here. I believe that all of these issues could be solved by adapting existing control settings like *FakeLab* or *ControlArena*.

Initial experiments attempted to use *GPT-6.1 Sol* as the agent, and *GPT-6 Luna* as the weaker trusted monitor, but were incorrectly flagged as a cybersecurity-related request and therefore failed. This, combined with the constraint of developing a custom environment and tasks in the allotted time meant that we used weaker models on task pairs that were less challenging and less malicious. We saw that the agents did not collaborate on most tasks and scored high secret task completion rates, which would be explained by insufficiently difficult tasks not requiring extensive teamwork, and hence making lots of communication more suspicious. Insufficient time and compute also limited experiments to very small numbers of agents, and so generalisation to larger multi-agent systems would have been unclear even with successful experiments.

References

J. Benton, M. Wagner, E. Christiansen, C. Anil, E. Perez, J. Srivastav, E. Durmus, D. Ganguli, S. Kravec, B. Shlegeris, J. Kaplan, H. Karnofsky, E. Hubinger, R. Grosse, S. R. Bowman, and D. Duvenaud. Sabotage evaluations for frontier models. *arXiv preprint arXiv:2410.21514*, 2024. URL https://arxiv.org/abs/2410.21514.

A. Bhatt, C. Rushing, A. Kaufman, T. Tracy, V. Georgiev, D. Matolcsi, A. Khan, and B. Shlegeris. Ctrl-Z: Controlling AI agents via resampling. *arXiv preprint arXiv:2504.10374*, 2025. URL https://arxiv.org/abs/2504.10374.

R. Greenblatt, B. Shlegeris, K. Sachan, and F. Roger. AI control: Improving safety despite intentional subversion. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, pages 16295–16336, 2024. URL https://proceedings.mlr.press/v235/greenblatt24a.html.

X. Gu, X. Zheng, T. Pang, C. Du, Q. Liu, Y. Wang, J. Jiang, and M. Lin. Agent Smith: A single image can jailbreak one million multimodal LLM agents exponentially fast. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, pages 16647–16672, 2024. URL https://proceedings.mlr.press/v235/gu24e.html.

R. Inglis, O. Matthews, T. Tracy, O. Makins, T. Catling, A. Cooper Stickland, R. Faber-Espensen, D. O'Connell, M. Heller, M. Brandao, A. Hanson, A. Mani, T. Korbak, J. Michelfeit, D. Bansal,

T. Bark, C. Canal, C. Griffin, M. Stehlik, J. Wang, and A. Cooney. ControlArena, 2025. URL https://github.com/UKGovernmentBEIS/control-arena. GitHub repository, UK AI Security Institute.

E. Jones, A. Dragan, and J. Steinhardt. Adversaries can misuse combinations of safe models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pages 28327–28349, 2025. URL https://proceedings.mlr.press/v267/jones25a.html.

J. Kitaoka, T. A. Pham, C. R. Giudice, and J. Merizian. Red-teaming multi-agent attacks for loss-of-control risks. *arXiv preprint arXiv:2610.24052*, 2026. URL https://arxiv.org/abs/2610.24052.

J. Kutasov, Y. Sun, P. Colognese, T. van der Weij, L. Petrini, C. B. C. Zhang, J. Hughes, X. Deng, H. Sleight, T. Tracy, B. Shlegeris, and J. Benton. SHADE-Arena: Evaluating sabotage and monitoring in LLM agents. *arXiv preprint arXiv:2506.15740*, 2025. URL https://arxiv.org/abs/2506.15740.

D. Lee and M. Tiwari. Prompt infection: LLM-to-LLM prompt injection within multi-agent systems. *arXiv preprint arXiv:2410.07283*, 2024. URL https://arxiv.org/abs/2410.07283.

P. Lütje. The mechanics of a swarm: A reproducible external reconstruction of an unintended agent-coordination episode on a third-party wiki. *arXiv preprint arXiv:2609.12748*, 2026. URL https://arxiv.org/abs/2609.12748.

O. Makins, O. Angelini, Z. Shams, and M. Phuong. Multi-agent AI control: Distributed attacks hamper per-instance monitors. *arXiv preprint arXiv:2607.07368*, 2026. URL https://arxiv.org/abs/2607.07368.

S. R. Motwani, M. Baranchuk, M. Strohmeier, V. Bolina, P. Torr, L. Hammond, and C. Schroeder de Witt. Secret collusion among AI agents: Multi-agent deception via steganography. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*, pages 73439–73486, 2024. doi: 10.52202/079017-2336.

J. Needham, G. Edkins, G. Pimpale, H. Bartsch, and M. Hobbhahn. Large language models often know when they are being evaluated. *arXiv preprint arXiv:2505.23836*, 2025. URL https://arxiv.org/abs/2505.23836.

N. Radev, L. Haas, B. Arnav, and P. Bernabeu-Pérez. The best-laid SCHEMEs: Coordinated sabotage and monitoring in multi-agent systems. *arXiv preprint arXiv:2605.29178*, 2026. URL https://arxiv.org/abs/2605.29178.

J. L. Rippin, S. C. Marshall, D. D. Africa, and C. Schroeder de Witt. Tool use enables undetectable steganography in multi-agent LLM systems. *arXiv preprint arXiv:2606.28425*, 2026. URL https://arxiv.org/abs/2606.28425.